

UC MATH 600 MATHEMATICS FOR COMPUTER SCIENCE · MODULE 06 · LECTURE NOTES

Counting

Reasoning that looks like arithmetic.

MODULE OUTCOMES

- Use the sum, product, and bijection rules.
- Count permutations and combinations correctly.
- Correct for overcounting, including inclusion-exclusion.
- Prove identities by double counting.
- Use the pigeonhole principle.

1 The basic rules

SUM RULE

disjoint choices add.
 $|A \text{ union } B| = |A| + |B|$ when A and B are disjoint. "or"

PRODUCT RULE

independent choices multiply.
 $|A \times B| = |A| * |B|$. "and then"

BIJECTION RULE

if you can biject A with B, then $|A| = |B|$.
 The most powerful of the four: count something ELSE that is easier, and exhibit the bijection.

DIVISION RULE

if every element of B is hit exactly k times by a map from A, then $|A| = k |B|$.
 This is the overcounting correction (section 2).

AND THE DISCIPLINE: say which rule you are using at each step. A counting argument that does not name its rules is the kind that double-counts silently.

Name the rule at each step — a counting argument that does not name its rules is the kind that **double-counts silently**, and counting errors are unusually hard to spot because the answer is a plausible number either way. **The bijection rule is the one worth reaching for**: most elegant counting arguments are a bijection to something already counted, and §3's double counting is the same instinct from the other side.

2 Permutations and combinations

	Order matters	Order does not matter
Without repetition	$n! / (n - k)!$ — permutations	$C(n, k) = n! / (k!(n-k)!)$ — combinations
With repetition	n^k	$C(n + k - 1, k)$ — stars and bars

Ask 'does order matter' and 'can things repeat' before reaching for a formula — **two questions, four cases, and almost every counting error is answering one of them wrong** rather than misremembering an expression. **Two questions select the cell; that is the whole method**, and it is worth writing both answers down explicitly on every problem until it becomes automatic.

AND $C(n, k)$ IS THE DIVISION RULE: COUNT THE ORDERED ONES, THEN DIVIDE BY THE OVERCOUNT

There are $n! / (n - k)!$ ordered selections of k items from n , and each unordered set of k is counted exactly $k!$ times — once for each ordering of its elements.

So divide by $k!$ — which is precisely the division rule of §1 — and gives $C(n, k) = n! / (k!(n-k)!)$ without any need to remember the expression.

Which means you never actually need the formula: count with order, work out the overcount factor, and divide — and that method handles cases the standard formula does not cover, which is where it earns its keep.

And it generalises directly: the number of arrangements of a word with repeated letters is the factorial of its length divided by the factorial of each letter's multiplicity — the same argument, applied once per repeated letter.

3 Double counting

COUNT ONE SET TWO WAYS, AND THE TWO ANSWERS ARE EQUAL — WHICH IS A PROOF

'The sum over k of $C(n,k)$ equals 2^n ' is **immediate** once you see it: **both sides count the subsets of an n -element set**, the left by grouping them according to size and the right by choosing in-or-out for each element.

And Pascal's identity, $C(n,k) = C(n-1,k-1) + C(n-1,k)$, is the same move: split the k -subsets according to whether they contain the last element, and count each group.

Which is a proof, and a better one than the algebraic verification — it explains *why* the identity holds rather than confirming that the two expressions happen to be equal, and it is usually shorter.

And it is the technique Project 2 asks for: one counting problem solved two ways, with the two answers shown equal algebraically — which is the clearest available demonstration that counting is reasoning rather than arithmetic.

4 Pigeonhole

- $n + 1$ objects in n boxes means some box contains at least two — which is obvious, and the difficulty is always choosing the boxes, never the principle.
- Generalised: n objects in k boxes means some box contains at least $\lceil n/k \rceil$, which is the form actually used in practice.
- 'Two people in London have exactly the same number of hairs on their heads' — the boxes are possible hair counts and the objects are people, and the entire argument is noticing that there are more people than possible counts. Nothing else happens.
- And it gives non-constructive existence proofs: you learn that something exists without any way to find it — which is Module 02 §4's second flavour of existence proof, and is philosophically interesting and practically frustrating.
- It is also how hash collisions are proved inevitable (more possible inputs than outputs), which is where CSCE 629 and CSCE 711 both use it. The difficulty is always choosing the boxes: the principle is one line, and the entire content of any pigeonhole argument is what you decided to count.

AND INCLUSION-EXCLUSION, WHICH IS THE SUM RULE REPAIRED

The sum rule requires disjointness (§1), and when the sets overlap you subtract the pairwise intersections, add back the triple intersections, and so on with alternating signs.

Which is exactly the correction for having counted the overlaps more than once — subtract the double-counts, notice you have now removed the triples too many times, add them back — **and the alternating signs are that correction iterated to the end.**

And it is used constantly in probability (Module 11 §2), where it is the identical formula with measures in place of counts, which is a good example of the same combinatorial fact wearing two hats.

So this module's one-sentence summary: **count carefully, name the rule, and check the overcount** — which is three habits rather than a set of formulas to memorise, and the habits survive problems the formulas do not cover.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapter 15 (free)	The whole module , free — and the four rules are stated exactly as §1 gives them.
MIT 6.042J, the counting lectures (free video)	§§1 to 3, with the bijection rule used repeatedly.
Graham, Knuth & Patashnik — Concrete Mathematics	§§2 and 3 far beyond this course — demanding, excellent, and the binomial coefficient chapter is the reference. Library copy.
Hammack — Book of Proof, chapter 3 (free)	§§1 and 2, free, with many worked examples.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the four rules** and give a problem using each.
- 2 **Solve one problem by bijection** to something easier.
- 3 **For ten problems, answer the two questions** before computing.
- 4 **Derive $C(n,k)$** from the division rule.
- 5 **Count the arrangements of a word with repeats**, from first principles.

- 6 **Prove the subset-sum identity** by double counting.
- 7 **Prove Pascal's identity** by double counting.
- 8 **Solve one problem two ways** and show the answers agree algebraically.
- 9 **Use pigeonhole** on three problems, naming the boxes each time.
- 10 **Apply inclusion-exclusion** to a three-set problem.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the four counting rules and what each is for.
- 2 Which is most powerful, and why?
- 3 What two questions select the formula?
- 4 Derive $C(n,k)$ from the division rule.
- 5 How does that generalise to repeated letters?
- 6 What is double counting, and why is it a proof?
- 7 Prove one identity by it.
- 8 State pigeonhole, generalised, and say where the difficulty is.
- 9 What kind of existence proof does it give?
- 10 State inclusion-exclusion and what it repairs.